Optimal Taxation with Imperfect Take-up

Dylan T. Moore

University of Hawai'i Economic Research Organization (UHERO)

Department of Economics

University of Hawai'i at Mānoa

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Abstract

This paper revisits classic results in optimal taxation by accounting for the ways in which imperfect take-up of social/tax benefits alter the efficiency and equity characteristics of redistributive technologies. In a series of stylized models, I explore several implications of imperfect takeup. First, I show that the presence of non-filers in the income tax base suggests commodity subsidies may be preferable to income tax-based transfers as a mechanism of redistribution, due to the passive nature of this form of redistribution. The imperfect targeting associated with these subsidies may also rationalize a system of differential commodity taxation/subsidization. Second, I when filing is an endogenous choice and responds to incentives to file, I show that the impact of non-filers on the optimal level of redistribution depends on whether or not the costs of filing are welfare-relevant: if they are, the impact of non-filing on redistribution is ambiguous; if not, the effect is negative. The paper concludes with a brief discussion of some other policy design implications of imperfect takeup: for the choosing amongst transfer programs with different takeup characteristics; and for state-level SNAP policy in the United States.

Modern transfer systems routinely fail to reach all eligible households. Administrative complexity, information frictions, and stigma generate imperfect take-up of welfare benefits. In the case of income tax benefit programs, some agents never file taxes, while others file but do not claim the benefits to which they are entitled. These lapses can alter both the equity and efficiency of redistributive policy, but standard optimal tax models do not account for this. An increasingly large body of empirical work on documents substantial non-claiming rates, and examines policies which may influence takeup decisions (e.g. Bhargava, 2015; Goldin, 2022; Ko & Moffit, 2024) and calls for theoretical frameworks that treat takeup as a design constraint rather than an incidental administrative detail.

This paper helps to fill this gap in the literature by exploring several extensions of the optimal tax thoery that account for the possibility that some agents will not receive the transfers to which they are entitled. First, I study joint commodity-and-income taxation when a fraction of households are non-filers. Non-filing blunts the redistributive reach of the income tax, generically rationalizing commodity *subsidies* and—even under Atkinson–Stiglitz preference assumptions—differential commodity subsidization/taxation.

Next, I consider a simple model of nonlinear income taxation where taxpayers with negative tax liability do not always claim their transfer income, and claiming decisions are endogenous. I show that the optimal nonlinear tax schedule is characterized by the usual ABC formula, but the social value of the marginal dollar of revenue is altered by the imperfection and endogeneity of takeup. Non-filing may thus imply that either more or less redistribution will place at the optimum than in the standard model, depending on the pattern of takeup, how takeup responds to the generosity of the transfer, and—crucially—whether takeup costs are welfare-relevant, introducing a normative ambiguity dimension to the analysis (Reck & Goldin, 2022).

Finally, I consider discuss several additional results pertaining to imperfect takeup of benefit programs using an MVPF-based framwork which builds on the earlier results of the paper. This includes a discussion about the version of the equity efficiency tradeoff involved in comparing two benefit programs with different efficiency and takeup characteristics. I also discuss the nuances of state-level policy making in the United States, where state policymaker have significant opportunities to aid their residents by pursuing policies that encourage takeup of federal programs.

The remainder of the paper is organized as follows. Section 1 characterizes optimal commodity and income taxes with exogenous non-filing. Section 2 introduces a simple model with endogenous take-up and explores how non-filers affect the optimal level of redistribution. Section 3 discusses the two further policy application using a stylized MVPF framework. I conclude by outlining directions for future work.

1 Exogenous Non-Filing, Subsidies, & Rate Differentiation

The economy consists of two sets of agents: filers and non-filers. Non-filers have a type indexed by $\theta \in \Theta$, and filers have a type indexed by $w \in \mathcal{W}$. The cumulative distribution of filers types is F_w and non-filer types is F_Θ. Let $\mu \in (0,1)$ be the share of non-filers in the economy.

Filers choose how much income to earn, and how much after-tax income to spend on a set of consumption goods. Non-filers have a fixed budget to spend on the same set of consumption goods. For simplicity of exposition, we will discuss the two-good case.

Filers Let $z(w) \geq 0$ be the taxable income a type w filer selects and let $x_1(w) \geq 0$ and $x_2(w) \geq 0$ be their consumption of the two commodities. The set of choices for a type w filer is, $(z(w), x_1(w), x_2(w))$, the solution to

$$\max_{z, x_1, x_2} \left\{ u\left(\phi\left(x_1, x_2\right), z; w\right) : \left(1+t_1\right) x_1 + \left(1+t_2\right) x_2 = z - T(z) + n(w) \right\}, \tag{1}$$

where $T(\cdot)$ is a nonlinear income tax schedule, t_i is the commodity tax rate applied to good $i \in \{1,2\}$, and $n(w)$ is some untaxed exogenous nonlabor income.

A type w filer's problem can be framed as a two-stage problem. In the second stage, the agent chooses commodity consumption conditional on their choice of after-tax income I. Let $\Phi(I)$ represent the indirect utility the agent gets from this second stage problem for a given value of after-tax income I:

$$\Phi(I) \equiv \max_{x_1, x_2} \left\{ \phi\left(x_1, x_2\right) : \left(1+t_1\right) x_1 + \left(1+t_2\right) x_2 = I \right\}. \tag{2}$$

In the first stage, the a type w filer chooses a value of taxable income $z(w)$ which maximizes their utility, anticipating how this choice impacts the value of the third stage problem. Let $V^F(w)$ represent the indirect utility a type w filer gets from this first stage problem:

$$V^F(w) \equiv \max_z \left\{ u\left(\Phi\left(z - T(z) + n(w)\right), z; w\right) \right\}. \tag{3}$$

Non-filers Let $x_1^N(\theta) \geq 0$ and $x_2^N(\theta) \geq 0$ be the consumption of the two commodities for a non-filer of type θ. All non-filers are assumed to earn no income, but rather finance their consumption using some untaxed nonlabor income $n^N(\theta)$. In this section, we shall assume $\left(x_1^N(w), x_2^N(w)\right)$, the solution to

$$\max_{x_1,x_2}\left\{\phi(x_1,x_2) : (1+t_1)\,x_1 + (1+t_2)\,x_2 = n^N(\theta)\right\}. \tag{4}$$

Notice, because the preferences of non-filers have been assumed to be identical to those of filers, we have that the indirect utility of a type θ nonfiler is

$$V^N(\theta) \equiv \Phi\left(n^N(\theta)\right).$$

Planner's Problem The social planner's problem is to select $T(\cdot)$, t_1, and t_2 to maximize the weighted social welfare function

$$(1-\mu)\int \gamma^F(w)\,V^F(w)\,\mathrm{d}F_w(w) + \mu\int \gamma^N(\theta)\,V^N(\theta)\,\mathrm{d}F_\theta(\theta)$$

subject to a budget constraint

$$(1-\mu)\left[\int T(z(w))\,\mathrm{d}F_w(w) + t_1\bar{x}_1^F + t_2\bar{x}_2^F\right] + \mu\left[t_1\bar{x}_1^N + t_2\bar{x}_2^N\right] = 0$$

where average demand for good i by filers is

$$\bar{x}_i^F \equiv \int x_i(w)\,\mathrm{d}F_w(w),$$

and average demand for good i by non-filers is

$$\bar{x}_i^N \equiv \int x_i^N(\theta)\,\mathrm{d}F_\theta(\theta).$$

1.1 Optimal Tax System with Homogeneous Non-labor Income

First, consider the case where $n(w) = n^N(\theta) = 0$. In this scenario, the social planner cannot do anything to affect non-filers. All non filers have no taxable income, receive no transfers, and don't buy any taxed commodities. Hence, the optimal tax system is really the solution to a modified planner's problem which excludes the non-filers' welfare from the objective function. Further note that in this special case, filer preferences satisfy the requirements of the Atkinson-Stiglitz theorem, hence our optimal tax system will consist of a standard ABC formula to characterize the optimal nonlinear income tax schedule, and a zero commodity tax rate.[1]

1.1.1 Positive Nonlabor Income

Next, consider the case where $n(w) = n^N(\theta) = n_0 > 0$. In this scenario, the social planner can improve non-filer welfare by subsidizing consumption. All non-filers have an identical income, which they use to buy the taxed commodities. If we subsidize those commodities, we can improve non-filer welfare.

Note that any type w filer such that $z(w) = 0$, will select the same bundle of commodities as a non-filer. This is because both groups have identical preferences over commodities, and identical incomes. Given this, the requirements of the Atkinson-Stiglitz theorem remain satisfied, so we know that optimal commodity taxes will be uniform.

Let us further assume that any such filers achieve the same level utility as non-filers ($V^F(w) = V^N(\theta)$) and that welfare weight assigned to these groups are the same ($\gamma^F(w) = \gamma^N(\theta)$). This is a natural assumption: why should the planner prefer one group of non-working individuals over another if both have identical incomes, simply because one group files a tax return?

Under this assumption, we can find the optimal tax system in two steps. First, set $t_1 = t_2 = 0$, and solve for the optimal schedule under the counterfactual assumption that all the non-filers are filers with types w such that they will always select to earn no income ($z(w) = 0$). This "optimal" schedule, $\tilde{T}(\cdot)$, follows the familiar ABC formula, and will implies some specific value for the "optimal" size of the demogrant: $\tilde{T}(0) < 0$.

In the second step, we'll adjust the tax schedule $\tilde{T}(\cdot)$ to implement an identical budget constraint for tax filers under a uniform commodity subsidy. Let t be the optimal uniform commodity tax rate and $T(\cdot)$ be the optimal nonlinear tax schedule. Note that a tax filer's budget constraint in this optimal system is

$$(1+t)\,x_1 + (1+t)\,x_2 = n_0 + z - T(z)\,.$$

As Feldstein (1999) notes, this is equivalent to the budget constraint that can be obtain under a zero commodity tax with a slightly adjusted tax schedule

$$\begin{aligned} x_1 + x_2 &= \frac{n_0 + z - T(z)}{1+t} \\ &= z + n_0 - \underbrace{\frac{T(z) + t\,(z + n_0)}{1+t}}_{\text{alternative tax schedule}} \end{aligned}$$

[1] In fact, the planner is indifferent between this tax system and other tax systems with a uniform commodity tax, with suitably modified income tax rates.

The optimal nonlinear tax schedule $T(\cdot)$ and uniform commodity tax rate t can be obtained by simply taking of tax schedule from step one, $\tilde{T}(\cdot)$, and asking what alternative tax system satisfies

$$\tilde{T}(z) = \frac{T(z) + t(z + n_0)}{1 + t},$$

ensuring all filers face equivalent budget constraints to the original schedule, and $T(0) = 0$, ensuring that a commodity tax is used as the instrument of redistribution rather than a demogrant. Taken together, these requirements imply that

$$T(z) = (1 + t)\tilde{T}(z) - t(z + n_0)$$

and

$$t = \frac{\tilde{T}(0)}{n_0 - \tilde{T}(0)} < 0.$$

This new system accomplishes what the step 1 system was trying to do, but could not because of non-filers do not receive the demogrant. The optimal tax system is thus simply a feasible way of implementing the planner's preferred allocation, given the constraints introduced by the non-filers.

While this model is very simple, it captures a critical aspect of optimal taxation with non-filers. In the standard model, under Atkinson-Stiglitz preferences, the planner is indifferent between a range of possible tax systems which combine uniform commodity taxation with a nonlinear income tax, but which all implement the same allocation. But in the model with non-filers, these different tax systems are no longer equivalent, and the planner has a preference for a specific nonlinear tax schedule and a specific uniform commodity subsidy.

This result has parallels in prior literature and in optimal tax folk wisdom. Informally, the Atkinson-Stiglitz theorem has often been taken to imply that a broad-based commodity tax is good policy, on the basis of enforcement/administrative considerations. A very different justification for resolving the planner's indifference amongst these systems can be found in Moore (2024) who discusses implications of tourist consumption for optimal taxation in a destination economy. The result of this section offers an alternative resolution, with very different policy implications: broad-based subsidies. To obtain comprehensive policy implications thus might require a model which incorporates both non-filers and these other considerations simultaneously. This is beyond the scope of this paper.

1.2 Optimal Tax System with Heterogeneous Non-labor Income

Let us assume that non-filer non-labor income is heterogeneous, while filer non-labor income is fixed at some value $n^F(w) = n_0$, so that filer preference still satisfy the assumptions that would otherwise result in Atkinson-Stiglitz theorem holding. Heterogeneity of non-filer non-labor income is sufficient to imply that a differentiated commodity tax/subsidy system is preferable to a uniform commodity tax subsidy when one of the two goods is a "necessity" or "luxury" good amongst the non-filers, such that differential commodity taxation allows the planner to better target redistribution to higher marginal utility non-filers (i.e. to put the subsidy on the good disproportionately consumed by the lower income non-filers).

To see why, consider tax reforms that jointly change the tax system in a way that is distribution neutral for filers, in the spirit of **?**. Suppose that the economy begins with the tax system $(t_1, t_2, T(\cdot))$. Now, consider a joint tax reform which marginally increases tax rate on good i by $\mathrm{d}t_i > 0$ while also marginally changing income tax liability at every income level z by some amount $\mathrm{d}T(z)$. The effect of this joint reform on the value of the filers' second stage problem (equation **??**) is:

$$\mathrm{d}\Phi\left(z - T(z) + n_0\right) = \frac{\partial \Phi\left(z - T(z) + n_0\right)}{\partial I}\left[x_i(z)\,\mathrm{d}t_i + \mathrm{d}T(z)\right]$$

Notice, by setting $\mathrm{d}T(z) = -x_i(z)\,\mathrm{d}t_i$ for all z, we obtain a joint reform for which $\mathrm{d}\Phi(z) = 0$ for all z.

This type of joint reform is distribution neutral from the perspective of filers, as the income tax change is designed to compensate every filer for the loss in purchasing power caused by the increase in the commodity tax rate on good i. Consequently, when we evaluate the first-order welfare effects of this joint reform, we do not need to account for effects on the private utility of residents, but only for effects on revenue obtained from residents. Further, note that because the reform leaves the value the filers' second stage problem unchanged at every value of z, it also leaves the taxable income choices of filers unchanged.

For non-filers, the first-order welfare of the joint reform is just the standard effects of a change to a linear commodity tax rate. Thus, the planner's FOC for this joint reform is

$$-\mu\left[\int \gamma^N(\theta)\,\frac{\partial \Phi(\theta)}{\partial n} x_i^N(\theta)\,\mathrm{d}F_\theta(\theta)\right] + \lambda\left.\frac{\mathrm{d}R}{\mathrm{d}t_i}\right|_{\mathrm{d}T(z) = -x_i(z)\mathrm{d}t_i} = 0$$

where the revenue effect is

$$\left.\frac{\mathrm{d}R}{\mathrm{d}t_i}\right|_{\mathrm{d}T(z)=-x_i(z)\mathrm{d}t_i} = \mu\bar{x}_i^N + \underbrace{\mu\int\left[t_i\left.\frac{\partial x_i^N(\theta)}{\partial t_i}\right|_u + t_{-i}\left.\frac{\partial x_{-i}^N(\theta)}{\partial t_i}\right|_u\right]\mathrm{d}F_\theta(\theta)}_{\text{nonfiler substitution effects}}$$
$$\underbrace{-\mu\int x_i^N(\theta)\left[t_1\frac{\partial x_1^N(\theta)}{\partial I} + t_2\frac{\partial x_2^N(\theta)}{\partial I}\right]\mathrm{d}F_\theta(\theta)}_{\text{nonfiler income effects}}$$
$$+\underbrace{(1-\mu)\int\left[t_i\left.\frac{\partial x_i(w)}{\partial t_i}\right|_{u,z=z(w)} + t_{-i}\left.\frac{\partial x_{-i}(w)}{\partial t_i}\right|_{u,z=z(w)}\right]\mathrm{d}F_w(w)}_{\text{filer substitution effects}} \quad (5)$$

Applying standard properties of Hicksian demand functions to the filers' compensated responses, we can simplify this to:[2]

$$\left.\frac{\mathrm{d}R}{\mathrm{d}t_i}\right|_{\mathrm{d}T(z)=-x_i(z)\mathrm{d}t_i} = \mu\bar{x}_i^N - \underbrace{\mu\left[\frac{t_i}{1+t_i} - \frac{t_{-i}}{1+t_{-i}}\right]\int x_i^N(\theta)\,\varepsilon_i^N(\theta)\,\mathrm{d}F_\theta(\theta)}_{\text{nonfiler substitution effects}}$$
$$\underbrace{-\mu\int x_i^N(\theta)\,\alpha(\theta)\,\mathrm{d}F_\theta(\theta)}_{\text{nonfiler income effects}}$$
$$-\underbrace{(1-\mu)\left[\frac{t_i}{1+t_i} - \frac{t_{-i}}{1+t_{-i}}\right]\int x_i(w)\,\varepsilon_{i|z}(w)\,\mathrm{d}F_w(w)}_{\text{filer substitution effects}}$$

where the compensated elasticity of demand for good i is

$$\varepsilon_{i|z}(w) \equiv -\frac{1+t_i}{x_i(w)}\left.\frac{\partial x_i(w)}{\partial t_i}\right|_{u,z}$$

for filers, is

$$\varepsilon_i^N(\theta) \equiv -\frac{1+t_i}{x_i^N(\theta)}\left.\frac{\partial x_i^N(\theta)}{\partial t_i}\right|_u$$

[2]Note that

$$(1+t_i)\left.\frac{\partial x_i(w)}{\partial t_i}\right|_{u,z} + (1+t_{-i})\left.\frac{\partial x_i(w)}{\partial t_{-i}}\right|_{u,z} = 0.$$

Combined with Slutsky symmetry, this implies that

$$\left.\frac{\partial x_{-i}(w)}{\partial t_i}\right|_{u,z} = -\frac{1+t_i}{1+t_{-i}}\left.\frac{\partial x_i(w)}{\partial t_i}\right|_{u,z}. \quad (6)$$

Similarly, non-filer compensated responses satisfy

$$\left.\frac{\partial x_{-i}^N(\theta)}{\partial t_i}\right|_u = -\frac{1+t_i}{1+t_{-i}}\left.\frac{\partial x_i^N(\theta)}{\partial t_i}\right|_u.$$

for nonfilers, and where the income effect for nonfilers is

$$\alpha(\theta) \equiv t_1 \frac{\partial x_1^N(\theta)}{\partial I} + t_2 \frac{\partial x_2^N(\theta)}{\partial I}.$$

The planner's FOC can thus be simplified to

$$1 - \bar{g} - \text{Cov}\left(\frac{x_i^N(\theta)}{\bar{x}_i^N}, g(\theta)\right) = \left[\frac{t_i}{1+t_i} - \frac{t_{-i}}{1+t_{-i}}\right]\left(\mathcal{E}_i^N + \frac{1-\mu}{\mu}\frac{\bar{x}_i^F}{\bar{x}_i^N}\mathcal{E}_{i|z}^F\right) \tag{7}$$

where the social marginal utility of income for nonfilers is

$$g(\theta) \equiv \frac{\gamma^N(\theta)\frac{\partial\Phi(\theta)}{\partial n}}{\lambda} + \alpha(\theta),$$

aggregate compensated demand elasticities for good i are

$$\mathcal{E}_i^N \equiv \frac{\int x_i^N(\theta)\,\varepsilon_i^N(\theta)\,\mathrm{d}F_\theta(\theta)}{\bar{x}_i^N}$$

and

$$\mathcal{E}_{i|z}^F \equiv \frac{\int x_i(w)\,\varepsilon_{i|z}(w)\,\mathrm{d}F_w(w)}{\bar{x}_i^F}.$$

In addition, note that the average social welfare weight for nonfilers is $\bar{g} = \int g(\theta)\,\mathrm{d}F_\theta(\theta)$ and the average nonfiler consumption of good i is $\bar{x}_i^N \equiv \int x_i^N(\theta)\,\mathrm{d}F_\theta(\theta)$.

Considering equation 7 for all $i \in \{1,2\}$, we can obtain an intuitive condition for the optimal level of commodity tax differentiation

$$\frac{t_1}{1+t_1} - \frac{t_2}{1+t_2} = \frac{-\text{Cov}\left(g(\theta), \frac{x_1^N(\theta)}{\bar{x}_1^N} - \frac{x_2^N(\theta)}{\bar{x}_2^N}\right)}{\mathcal{E}_1^N + \mathcal{E}_2^N + \frac{1-\mu}{\mu}\left(\frac{\bar{x}_1^F}{\bar{x}_1^N}\mathcal{E}_{1|z} + \frac{\bar{x}_2^F}{\bar{x}_2^N}\mathcal{E}_{2|z}\right)}. \tag{8}$$

This condition has the intuitive implication that the tax rate on good 1 should be higher if—amongst nonfilers—"relatively high consumption" of good 1 is negatively correlated with social marginal utility of income.

Note, the definition of "relatively high consumption" is somewhat subtle here. The question is not whether households with higher social welfare weights spend a lower share of their income on good 1, but rather whether they tend to have a lower value of $\frac{x_1^N(\theta)}{\bar{x}_1^N} - \frac{x_2^N(\theta)}{\bar{x}_2^N}$: the difference between normalized consumption of good 1 and 2 (relative to the average consumption of each good). Importantly, variation in consumption patterns amongst the filers plays no role in equation 8, and properties of filer demand do not determine whether tax rate differentiation is optimal, or the direction of optimal differentiation. Filer demand only plays a role in the optimal magnitude of tax rate differentiation.

Ultimately, the result is quite straightforward. Amongst filers, all redistribution can be conducted using the nonlinear income tax. Redistribution from filers to nonfilers can be conducted using the same type of tax structure discussed in the homogeneous nonlabor income case: a uniform commodity subsidy. But when non-labor income is heterogeneous amongst non-filers, the planner will also want to redistribute amongst nonfilers. Commodity tax differentiation is the only way they can accomplish this.

However, the equity benefits of commodity tax differentiation must be weighed against the efficiency costs that differentiation creates. The denominator on the righthand side of 8 measures the magnitude of these costs. The magnitude of optimal differentiation declines the greater are these costs. This attenuation is also increasing in the ratio of filer to nonfiler average demand for each good—$\frac{\bar{x}_1^F}{\bar{x}_1^N}$ and $\frac{\bar{x}_2^F}{\bar{x}_2^N}$—and decreasing in the nonfiler share μ. This happens because the equity benefits of differentiation are exclusive to the nonfiler population, so the larger the relative size of filer consumption, the more weight is placed of the efficiency costs of differentiation stemming from filers.

1.3 Generalizing the Results

Give the important role of non-labor income in the results above, one may be tempted to question their policy relevance. The critical requirement for the planner to be able to redistribute to non-filers using commodity subsidies is that non-filers must have some purchasing power. For simplicity, the results above are derived in a model where non-filers all have no labor income, but in reality, this is not the case. The inclusion of non-labor income is necessary to obtain interesting results in this simplified setting.

The results do generalize beyond this extreme special case. In more general models allowing for non-filers who earn income, modest generalizations of these results still obtain, however in such models the planner will generally no longer find it optimal to set the demogrant equal to zero. Rather, they rely on a combination of the demogrant and commodity subsidies (targeted to certain goods) for their redistributive aims. Still, the question remains: can commodity subsidies be used to redistribute to non-filers who earn no income? To answer yes, we may need to look beyond the individual non-filer themselves. For example, commodity subsidies might make it cheaper for charitable organization to provide support to these zero income non-filers.

2 Endogenous Filing & Optimal Redistribution

Let us consider a different simple model from the one discussed in section 2. Consider an economy with two groups of income taxpayers: *always filers* and, *potential non-filers*. As in section 1, these taxpayers are subject to a nonlinear income tax schedule which is used to finance a demogrant, but only those who file their taxes receive the demogrant. Further suppose that

(i) the planner cannot levy commodity taxes, only an income tax;

(ii) all the *always filers* file their taxes, and thus receive the demogrant;

(iii) only some *potential non-filers* fail to file, but some actually do file their taxes;

(iv) the share of *potential non-filers* who file is increasing in the incentive to file;

(v) and, all *potential non-filers* earn zero income, whereas only some *always filers* do so.

In this model, the incentive to file taxes stems from the promise of receiving demogrant, G. Potential non-filers who file their taxes do not change their labor supply, continuing to earn nothing, but will have higher consumption than those who do not file because they receive a demogrant. Absent a cost of filing, their utility would thus be higher upon filing.

Let the indirect utility of type w always filers be

$$V^F(w) \equiv \max_z \{u(z - T(z) + G, z; w)\},$$

and let their choice of income be $z(w)$. Further suppose that always filers with a type of zero ($w = 0$) have no income ($z(0) = 0$), whereas filers with a positive type ($w > 0$) always have positive income.

Let the indirect utility of type θ potential non-filers who file be

$$V^{NF}(\theta) \equiv V^N(G; \theta).$$

Let the indirect utility of type θ potential non-filers who do not file be

$$V^{NN}(\theta) \equiv V^N(0; \theta) + \rho\xi$$

where ξ is the cost of filing, and $\rho \in [0, 1]$ is a parameter which determines whether the cost of filing is *normatively relevant* to the social planner's calculus. Varying ρ will allow us to consider the differing implications of assuming non-filing is a rational response to the cost of filing, or of assuming it is a mistake of the agent.

Regardless of our normative assumptions about the cost of filing, we assume that it does influence agent behavior. We assume that each type θ potential non-filers draws a cost of filing ξ from some cost distribution and only files their tax return if

$$V^N(G;\theta) - V^N(0;\theta) > \xi.$$

We then let $\pi(G;\theta)$ denote the probability a type θ potential non-filers files their tax return. If we assume that $V^N(G;\theta)$ is strictly increasing in G, and this implies that $\pi(G;\theta)$ is increasing in G. We further denote the aggregate share of potential non-filers who file by

$$\Pi \equiv \int \pi(G;\theta)\,\mathrm{d}F_\theta(\theta).$$

Let the type distribution amongst *positive income always filers* be F_w, denote their average welfare by

$$W^{F+} \equiv \int \gamma^F(w)\,V^F(w)\,\mathrm{d}F_w(w).$$

Let the *zero income non-filers* be homogeneous, with an average welfare denoted by

$$W^{F0} \equiv \gamma_0^F V_0^F.$$

Let the type distribution amongst *potential non-filers* be F_θ, with the average welfare of those who file denoted by

$$W^{NF} \equiv \frac{\int \gamma^N(\theta)\,\pi(G;\theta)\,V^{NF}(\theta)\,\mathrm{d}F_\theta(\theta)}{\Pi},$$

and the average welfare of those who fail to file denoted by

$$W^{NN} \equiv \frac{\int \gamma^N(\theta)\,(1-\pi(G;\theta))\,V^{NN}(\theta)\,\mathrm{d}F_\theta(\theta)}{1-\Pi}.$$

Let $\alpha \in (0,1)$ denote the share of agents in the economy with zero labor income. Further, let $\mu \in (0,1)$ be the share of zero labor income agents that are potential non-filers. Given these definitions, we can characterize the population share of each of our four groups of taxpayers: (i) $1-\alpha$ are positive income always filers; (ii) $\alpha(1-\mu)$ are zero income always filers; (iii) $\alpha\mu\Pi$ are potential non-filers who choose file their taxes; and, (iv) $\alpha\mu(1-\Pi)$ are potential non-filers who fail to file. In our discussion of comparative statics later in this section, it will prove helpful to have set up the model in such a way that as μ varies, the share of zero income agents does not change.

The social planner's problem is to select a non-linear income tax schedule $T(\cdot)$ to maximize the weighted social welfare function

$$(1-\alpha) W^{F+} \quad + \quad \alpha(1-\mu) W^{F0} \quad + \quad \alpha\mu\left[\Pi W^{NF} + (1-\Pi) W^{NN}\right],$$

subject to a budget constraint

$$(1-\alpha)\int T(z(w))\,\mathrm{d}F_w(w) = G\left[1-\alpha\mu(1-\Pi)\right].$$

It is straightforward to show that in this model, that the optimal marginal tax rate schedule follows the standard ABC formula. For simplicity, we shall assume no income effects, and thus obtain the following characterization:

$$\frac{T'(z(w))}{1-T'(z(w))} = \frac{1}{\varepsilon(w)}\frac{1-F_w(w)}{wf_w(w)}\int_w^\infty\left(1-\frac{\gamma^F(u)}{\lambda}\frac{\partial V^F(u)}{\partial c}\right)\frac{\mathrm{d}F_w(u)}{1-F_w(w)} \tag{9}$$

where $\varepsilon(w)$ is the "virtual" elasticity of taxable income (i.e. not accounting for the way that the empirical ETI is modified by the curvature of the tax schedule). We complete the characterization of the tax schedule by imposing $T(0)=0$, since the demogrant is playing the role of the intercept term here.

Notice, the influence that zero income agents have on the characterization above is through the Lagrange multiplier λ, which measures the social value of relaxing the planner's budget constraint. We can see this clearly by considering the planner's FOC with respect to the demogrant, which can be written as

$$\lambda = \frac{(1-\alpha) W_G^{F+} + \alpha\left[(1-\mu) W_G^{F0} + \mu\Pi W_G^{NF} + \rho\mu\Pi_G\Delta\right]}{1-\alpha\mu(1-\Pi)+\alpha\mu G\Pi_G}$$

where f_G denotes the derivative of some function f with respect to G, and Δ measures the average increase in utility from filing for the set of potential non-filers who are at the margin of filing, under the assumption that filing costs are not welfare relevant. In particular,

$$\Delta \equiv \frac{\int \gamma^N(\theta)\left[V^{NF}(\theta)-V^{NN}(\theta)\right]\frac{\partial\pi(G;\theta)}{\partial G}\mathrm{d}F_\theta(\theta)}{\int\frac{\partial\pi(G;\theta)}{\partial G}\mathrm{d}F_\theta(\theta)}.$$

Suppose we begin at a tax system $(G, T(\cdot))$ which is initially optimal, and ask how the optimal system changes as the share of potential non-filers (μ) rises. Holding constant the tax system, if $\frac{\partial\lambda}{\partial\mu} > 0$, then equation (9) implies that marginal tax rates at all levels—and, consequently, total tax revenue—must rise in order to satisfy the optimality condition (9).[3]

Suppose that the average welfare of any group of zero income filers is the same, irrespective of whether they are always filers or potential non-filers who file taxes. More formally, suppose that $W^{NF} = W^{F0} = W^0$. This is a reasonable normative assumption if no one in this group has any nonlabor income outside of the demogrant, since then agents in both groups have the same consumption, labor supply, and filing decsion. In this case, it can be shown that[4]

$$\frac{\partial\lambda}{\partial\mu} > 0 \iff \underbrace{(1-\Pi)(1-\alpha)\left(W_G^{F+} - W_G^0\right)}_{\text{exogenous non-filer effect } (<0)} + \underbrace{G\Pi_G\left[\rho\frac{\Delta}{G} - (1-\alpha)W_G^{F+} - \alpha W_G^0\right]}_{\text{endogenous non-filer effect}} > 0. \tag{10}$$

To unpack the intuition here a bit further, consider the case where $\Pi_G = 0$, so that all non-filing is exogenous. In this case, condition (10) implies that increasing the share of potential non-filers decreases the optimal level of redistribution, as long as the marginal social value of the demogrant is higher for zero income filers than for positive income filers ($W_G^{F+} < W_G^0$), which will always be the case for standard, anonymous social welfare functions. The reason for this result is straightforward: with exogenous filing, the potential non-filers who don't file are effectively irrelevant to the planner's problem, thus as μ rises, the effective relative size of the zero income population is decreasing. Put more simply, if there are some agents who are totally beyond the reach, less redistribution occurs at the optimum because the planner does not need to raise money for redistribution to unreachable agents. Importantly, this does not imply that the optimal size of the demogrant G will fall: rather, the total revenue extracted for the income earning population falls. However, this smaller revenue amount is also being divided up amongst a smaller effective population of agents, since the non-filers are excluded.

Next, consider the case where non-filing is endogenous ($\Pi_G > 0$), but the cost of filing is fully welfare-relevant ($\rho = 0$). This means that the envelope theorem applies to any marginal non-filers who are induced to file by an increase in the demogrant: any gains are fully offset by the cost of filing. In other words, there is no corrective role for policy here. Unsurprisingly, in this scenario, endogenous filing effects serve to reinforce the

[3]Because the term measuring the average relative social marginal utility of income for types above w

$$\int_w^\infty \left(1 - \frac{\gamma^F(u)}{\lambda}\frac{\partial V^F(u)}{\partial c}\right)\frac{\mathrm{d}F_w(u)}{1-F_w(w)}$$

is increase in λ.

[4]

$$\frac{\partial\lambda}{\partial\mu} = \alpha\frac{G\Pi_G\left(\frac{\rho\Delta}{G} - (1-\alpha)W_G^{F+} - \alpha W_G^0\right) + (1-\alpha)\left(W_G^{F+} - W_G^0\right)(1-\Pi)}{\left(1-\alpha\mu(1-\Pi) + \alpha\mu G\Pi_G\right)^2}$$

negative impact of the share of potential non-filers on the level of redistribution, because in this scenario, endogenous non-filing simply serves to create a fiscal externality associated with raising the demogrant. Note, absent income effects, the marginal value of public funds (MVPF) of a demogrant in this model is

$$MVPF_G \equiv \frac{1}{1+\frac{\alpha\mu G\Pi_G}{1-\alpha\mu(1-\Pi)}}$$

The increase in the efficiency costs of redistribution associated with a higher share of potential non-filers thus lead to lower level of optimal redistribution.

Finally, consider the case where $\rho = 1$, so that filing costs are not welfare-relevant. In this case, non-filers are making a mistake from the planner's perspective. Non-filing behavior is creating an internality, introducing a corrective role for the demogrant. It is possible for this corrective value to be sufficiently large that the optimal level of redistribution is increasing in the share of potential non-filers. To see this, note that the term capturing the corrective motive can be written as

$$\frac{\Delta}{G} = \frac{\int \gamma^N(\theta)\left[\frac{1}{G}\int_0^G \frac{\partial V^N(g;\theta)}{\partial g}\mathrm{d}g\right]\frac{\partial \pi(G;\theta)}{\partial G}\mathrm{d}F_\theta(\theta)}{\int \frac{\partial \pi(G;\theta)}{\partial G}\mathrm{d}F_\theta(\theta)} \tag{11}$$

and further note that, under standard assumptions about the diminishing marginal social value of consumption, we will have

$$\frac{\partial V^{N,F}(G;\theta)}{\partial G} = \frac{\partial V^N(G;\theta)}{\partial G} < \frac{1}{G}\int_0^G \frac{\partial V^N(g;\theta)}{\partial g}\mathrm{d}g.$$

If $\gamma^N(\theta)\frac{\partial V^N(G;\theta)}{\partial G}$ is constant in θ, this implies that the *endogenous non-filer effect* term in equation (10) is always positive, since then we have

$$\frac{\Delta}{G} > W_G^{F0} > W_G^{F+} \implies \frac{\Delta}{G} > (1-\alpha)W_G^{F+} + \alpha W_G^{F0}$$

Whether the optimal level of redistribution is increasing or decreasing in the share of potential non-filers depends on whether it is sufficiently positive to outweigh the negative *exogenous non-filer effect.* In general, either outcome is possible. Condition (10) implies that a more responsive the filing rate (to increases in the demogrant), favors a positive effect on the level of redistribution.

However, it is also important to consider the case where $\gamma^N(\theta)\frac{\partial V^N(G;\theta)}{\partial G}$ varies across non-filers. As can be seen in equation (11), the density of type θ agents amongst marginal filers is $\frac{\frac{\partial \pi(G;\theta)}{\partial G}f_\theta(\theta)}{\int \frac{\partial \pi(G;\theta)}{\partial G}f_\theta(\theta)\mathrm{d}\theta}$, whereas the density of type θ agents amongst potential non-filers who file their taxes is $\frac{\pi(G;\theta)f_\theta(\theta)}{\int \pi(G;\theta)f_\theta(\theta)\mathrm{d}\theta}$. This difference matters a lot in scenarios where the social marginal value of consumption varies with type. For example, suppose the non-filing decisions exhibit a kind of self-targeting pattern, wherein those agents with the highest social marginal utility of income are the ones most likely to file. This could arise if, for example, some non-filers have non-labor income which lessens their need for the demogrant, making them less likely to file. In such a case, it is entirely possible in principle that the social marginal utility of consumption for the marginal non-filers could be low enough to reverse the sign of the endogenous non-filing effect, though this only occurs in the extreme case that their social marginal utility of consumption sinks below a weighted average of *always filer* social marginal utility of consumption,

$$\frac{\Delta}{G} < (1-\alpha)\, W_G^{F+} + \alpha W_G^{F0},$$

an average which includes all the filers that have positive labor income.

Nonetheless, we would generally expect the self-targeting to reduce the magnitude of the endogenous non-filer effect. By contrast, it seems altogether plausible that the opposite situation could emerge, where marginal non-filers are disproportionately individuals with high social marginal utility of consumption. This might happen for example, if nonlabor income is positively correlated with cognitive ability, with some attention budget, or with information. Or it might happen in cases where agents with low nonlabor income have a lower demand for tax filing services. To the extent that such a scenario holds, the endogenous non-filer effect becomes even more positive.[5]

To summarize, in a model with endogenous filing, as non-filers become a more important, this may lead the planner to favor a higher demogrant, in order to encourage more (potential) non-filers to file their taxes. That is to say, endogenous non-filing means that the demogrant is useful not only to increase the welfare of current filers, but also to extend redistribution to additional non-filers by getting them to file. However, this is only the case if there is a corrective role for government policy with respect to non-filing behavior. Furthermore, the importance of this corrective effect depends on the relative "deservingness" of the agents who would be induced to file; in the presence of self-targeting, the case for corrective action is somewhat diminished. In the opposite scenario, it may be enhanced.

[5]Note that earlier in this section, we made the simplifying assumption that all zero income filers have the same average welfare function:

$$W^{NF} = W^{F0} = W^0.$$

This assumption may no longer be tenable with heterogeneous non-filers of the kinds discussed above. If we relax this assumption, then $\frac{\partial\lambda}{\partial\mu} > 0$ if and only if

$$(1-\Pi)(1-\alpha)\left(W_G^{F+} - W_G^{F0}\right) + G\Pi_G\left[\frac{\rho\Delta}{G} - (1-\alpha)\, W_G^{F+} - \alpha W_G^{F0}\right] + \underbrace{\Pi\left(W_G^{NF} - W_G^{F0}\right)}_{\text{heterogeneity effect}} > 0.$$

Relative to condition (10), this condition requires us to account for any discrepancies in the social marginal utility of consumption between our two groups of zero income non-filers. It is not clear that this effect has a presumptive direction in any of the scenario discussed above, absent additional assumptions.

On the other hand, non-filing also implies that some taxpayers are beyond the planner's reach, rendering them effectively irrelevant to policy decisions, and there is no need to raise revenue to fund a demogrant for agents that cannot receive it. In addition, endogenous non-filing introduces a new type of fiscal externality, making it more costly to increase the demogrant than it otherwise would be. These competing effects together determine how the optimal level of redistribution is impacted by the introduction of non-filers.

While the simple model introduced above is only illustrative, it nonetheless serves to introduce key sufficient statistics which are also present in generalization of the mdoel. Critically, in a model with endogenous income tax filing, the optimal level of redistribution depends on the non-filing rate, Π and its responsiveness to the demogrant, Π_G, and on how the rate and responsiveness varies across different types of potential non-filers. If endogenous filing responses are higher amongst non-filers with relatively high social marginal utility of income, this favors a more redistributive optimal schedule. If the opposite is true, this will attenuate the optimal level of redistribution. Thus, an ideal empirical application of this framework would require information about not only responsiveness of filing to the incentive to file, but also how that responsiveness varies with consumption or wealth (some measure of differences in baseline welfare across households).

3 Further Implications of Imperfect Takeup

In this section, we depart from the consideration of fully worked optimal tax theory models to briefly discuss two other key implications of imperfect takeup for policy design, each of which builds on the core ideas illustrated in the examples above.

3.1 Where to Pour Okun's Bucket?

Okun (1975) presented the memorable metaphor for the equity-efficiency tradeoff by asking readers to consider a social planner who had to conduct redistribution by carrying money from the rich to the poor using a leaky bucket. Consider a slight twist on this story. Our planner would likely prefer to use a more centralized method of for distributing the funds collected in the bucket. Suppose the planner pours the money into a set of pipes, which carry the funds to the desired recipients. Policy analysis that ignores imperfect takeup might consider the possibility that the pipes are leaky—representing administrative costs, or fiscal externalities associated with redistribution. What is missing then, is consideration of the possibility that the pipes might not reach everyone the planner wants to get funds to: that the pipes leading to some folks are blocked, or were never built in the first place.

Consider a planner dealing with such imperfect systems of pipes, who has long employed an aging, and very leaky set of pipes that nonetheless successfully reach all households. Suppose a new set of pipes are introduced that promise to be nearly leak-free but, unlike the older pipes, do not reach all households, perhaps because the new pipes are not well suited to all areas where households live. When the planner has filled up Okun's bucket, they must make a decision: which set of pipes should we pour the bucket into? Using the old leaky pipes is, in a meaningful sense, quite wasteful. And yet, complete reliance on the new pipes would present an incredibly strong case for using the old ones if it means leaving some of the households unreached by the new pipes to suffer in a state of complete destitution.

We can formalize this variant of the equity-efficiency tradeoff using Hendren & Sprung-Keyser's (2020) framework. Spending more on a low efficiency, but broadly distributed policy B is preferable to spending more on high effiency, but narrowly distributed policy E if and only if

$$\gamma_B MVPF_B > \gamma_E MVPF_E$$

where $MVPF_i$ is the marginal value of public funds for policy i and γ_i is the social value placed on delivering a dollar of benefit to the beneficiaries of policy i. Suppose that policy B is a demogrant benefiting all households, but has a some marginal cost of administration per dollar transferred, $A_B \in (0, 1)$. By contrast, policy E only benefits a subgroup of households, but does so with zero administrative costs.

In this case, we have

$$MVPF_E = 1 > \frac{1}{1 + A_B} = MVPF_B$$

and

$$\gamma_B = (1 - \alpha)\,\gamma_0 + \alpha\gamma_E$$

where α is the share of households that do not benefit from policy E, due to imperfect takeup. The planner favors spending more on B if and only if

$$(1 - \alpha)\left(\frac{\gamma_0}{\gamma_E} - 1\right) > A_B$$

If the planner forgoes all spending on B in favor of E, then standard social welfare functions would lead us to expect $\frac{\gamma_0}{\gamma_E} > 1$, since the unreached households do not receive any transfers. If the ratio $\frac{\gamma_0}{\gamma_E}$ and the unreached population share α are large enough relative to the marginal administrative costs of policy B, it will lead the planner to favor increasing spending on B. That is to say, a planner who values the welfare of all households will—in certain cases—be willing to accept the use of a redistributive technology with greater efficiency costs if it has a wider reach.

The exception to this would be the case where unless there is significant self-targeting in the patterns of takeup for policy E. If the self-targeting is sufficiently important, it could well be the case that $\frac{\gamma_0}{\gamma_E} < 1$, resulting in an unambiguous preference for spending more on policy E. Even if this is not the case, the case for policy B is directly proportional to the ratio of welfare weights $\frac{\gamma_0}{\gamma_E}$. Self-targeting undermines this case by attenuating the ratio.

Endogeneity of takeup in the case of policy E alters the comparison of these two policies further. As we saw in the previous section, if greater spending on E increases takeup, this increases the net cost of delivering benefits via E due to the fiscal externality associated with the increased takeup. At the same time, it may introduce corrective benefits of E if imperfect takeup creates an internality. It also augments the interpretation of the welfare weights somewhat. Let b_E be the size of the benefit policy E delivers and let $\alpha(b_E)$ be the takeup rate as a function of the benefit. Further, note that γ_E is now interpreted as the welfare weight for inframarginal beneficiaries (who already takeup the benefit), whereas γ_E^M is the welfare weight for marginal beneficiaries (those induced to takeup the benefit). Finally, let Δ^M represent the net benefit to marginal beneficiaries of taking up the benefit, inclusive of any normatively relevant costs of takeup. Then the social value of increasing spending on

$$\gamma_E MVPF_E = \frac{(1-\alpha)\,\gamma_E + \alpha\sigma\gamma_E^M \frac{\Delta^M}{b_E}}{1-\alpha+\alpha\sigma},$$

where $\sigma(b_E) \equiv \frac{\alpha'(b_E) b_E}{\alpha(b_E)} > 0$, and I have suppressed the dependence of various terms on b_E. Spending more on policy B is preferred if and only if

$$(1-\alpha)\left(\frac{\gamma_0}{\gamma_E} - 1\right) > A_B + \underbrace{\frac{(1+A_B)\,\alpha\sigma}{1-\alpha+\alpha\sigma}\left(\frac{\gamma_E^M}{\gamma_E}\frac{\Delta^M}{b_E} - 1\right)}_{\text{endogeneity effect}}$$

Absent a corrective motive ($\Delta^M = 0$), the endogeneity effect is negative, and endogenous takeup ($\sigma > 0$) of E thus decreases the relative attractiveness of E. By contrast, in the presence of a corrective motive, the effect of endogeneity is more ambiguous. This effect only decreases the relative attractiveness of B if $\frac{\gamma_E^M}{\gamma_E}\frac{\Delta^M}{b_E} > 1$. This condition simply states that the social value of the benefits delivered to marginal beneficiaries $\gamma_E^M \frac{\Delta^M}{b_E}$ must have a value exceeding that of the benefits delivered to inframarginal beneficiaries. This depends in part on the magnitude of the corrective benefits of inducing marginal beneficiaries to claim the benefit, and on the relative social value placed on delivering benefits to marginal beneficiaries ($\frac{\gamma_E^M}{\gamma_E}$). As above, self-targeting with respect to E should be expected to attenuate the ratio $\frac{\gamma_E^M}{\gamma_E}$, thus diminishing the corrective value of investing in E.

3.2 State Policies Encouraging Federal Takeup

Rather than focusing on increasing takeup of a benefit by making the benefit larger, the planner might instead wish to focus on policies reducing the cost of takeup in some way, through things like providing information, streamlining application processes, or providing application assistance. A simple presentation of the MVPF associated with the federal government pursuing such a policy for a federally funded benefit is:

$$MVPF_c(c) = \frac{\alpha'(c)\, b}{1 + \alpha'(c)(b + A)}$$

where c is total spending on the takeup-inducing program, $\alpha(c)$ is the takeup rate as a function of c, b is the amount of the federally funded benefit per beneficiary, and A is the marginal administrative cost to the government associated with each beneficiary. This type of MVPF is not new. For example, Finkelstein and Notowidigdo (2019) apply a very similar formula to the results of an RCT examining programs which encourage SNAP takeup amongst elderly households. In one case, they estimate an MVPF of 0.89 for a treatment that provides elderly non-claimants of SNAP with information about their potential eligibility. However, as the authors themselves note, the perspective of the state government may differ. In general, when a benefit is partially federally funded the MVPF of a policy encouraging takeup from the perspective of the state government is

$$MVPF_c^s(c) = \frac{\alpha'(c)\, b}{1 + \alpha'(c)(\theta_b^s b + \theta_A^s A)}$$

where θ_b^s is the state's share of benefit costs and θ_A^s is the state's share of marginal administrative costs. In the particular case of SNAP, this reduces to

$$MVPF_c^s(c) = \frac{\alpha'(c)\, b}{1 + \alpha'(c)\frac{A}{2}},$$

since SNAP benefits are paid for by the federal government ($\theta_b^s = 0$) with states paying around half of administrative costs ($\theta_A^s = \frac{1}{2}$). Revisiting Finkelstein and Notowidigdo (2019)'s MVPF estimates for the information provision program and recalculating them from the state's perspective, one finds an MVPF of 10.65, even when conservatively assuming states pay for 100% of the marginal administrative costs.[6] Thus, a program to encourage takeup may look substantially more attractive from the perspective of a state

[6] As Agrawal, Hoyt, & Ly (2024), other discrepancies exist between MVPF calculations done from the perspective of a federal government verus a state government. For example, state policy changes may induce interstate migration effects, potentially resulting in fiscal, congestion, and pecuniary externalities.

government, as long as the program in question is (largely) paid for by the federal government.

This issue is not confined to SNAP policy alone. For example, state governments may find that programs to encourage their low income residents to claim the federal Earned Income Tax Credit (EITC), could be relatively attractive, even if they would be lacklustre if viewed from the perspective of the federal government. States might want to weigh such policies against other ways of helping these residents, such as implementing a state-level EITC top-up for example. However, such comparisons require careful consideration of the federal takeup effect of *both* policies: as Neumark & Williams (2020) suggest, it may well be the case that state EITCs themselves serve to increase takeup of the federal credit.

More generally, the possibility of spillovers from state benefit design to federal benefit takeup can radically alter the principles of good policy design for states. Consider for instance, a state policymaker contemplating the creation of a state program which complements federally funded SNAP benefits with a state financed program design to effectively eliminate the so called "SNAP cliff": a notch induced by the federal SNAP benefit schedule. At first glance, we might not expect such a program to be particularly attractive to the state. While it would help to eliminate a large earnings disincentive created by the cliff, this would come at great cost to the state, as those households which were bunching above the cliff increase to an earnings level above the cliff, causing the loss of a large federal benefit and the addition of a similarly large (though slightly smaller) state-funded benefit. The result is a situation where the the cost to the state from the movements of these former bunchers greatly exceeds the benefits created by such movements. While other positive fiscal externalities due to things like increased state income tax payments from former bunchers, or general equilibrium effects may attenuate the net cost, we might also expect some negative fiscal externalities due to the increased effective marginal rates above the (eliminated) notch.

This perspective misses a key consideration: how such a state program would impact takeup of federally funded SNAP benefits. Examining previous expansions of the SNAP eligible population which were implemented by shifting the cliff upwards, Anders & Rafkin (2024) find that such expansions of the SNAP eligible population lead to an average increase of 0.9 benefit enrollments amongst the previous eligible population for every one enrollment of a newly eligible household. Taken at face value, such a positive takeup effect could substantially changes the cost-benefit analysis of a supplementary state program to eliminate the cliff, because the federal government pays almost all the costs associated with the positive takeup response amongst the previously eligible group,. It seems quite plausible that similar effects would occur in the case of such a program, as long as it was administered jointly with the federal program in some capacity, such that there are no additional informational or compliance barriers associated with the supplementary program.

More generally, these examples suggest states should favor programs that parallel federal programs in various aspects of their design relative to creating new programs from scratch.